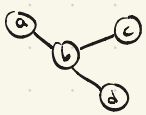
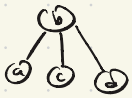
Def: A graph G consists of a pair of sets $V(G)$ and $E(G)$ where $V(G)$ is a finite non-empty set of objects called vertices and $E(G)$ is a set of unordered pairs of vertices called edges

Ex. Consider a graph G defined by $V(G) = \{a, b, c, d\}$ and $E(G) = \{ \{a,b\}, \{b,c\}, \{b,d\} \}$

This can be represented by:





Terminology:

- u, v are said to be adjacent if $\{u, v\}$ is an edge.
- vertices adjacent to a fixed vertex u are called neighbours of u. the set of all neighbours of u is defined $N_G(u)$
- an edge $e = \{u, v\}$ is incident with its two ends u and v.

Course specific info:

- $e = uv$ can also denote the edge $e = \{u, v\}$
- Edges are unordered; $uv = vu$
- For most of the course, we consider simple graphs: no loops, no multiple edges

Loop: 

Multiple edges: 

Graphs which allow loops and multiple edges are called multigraphs.

- we consider only finite graphs.

Graphs have many applications. Physical networks such as computer networks, electrical networks, road networks can be represented. They can be more abstract as well, such as social networks. Finally they are also used as data structures in computer science.

Def: the degree of a vertex v in a graph G is the number of edges incident with v. We write $\deg_G(v)$ or just $\deg(v)$ if G is understood. (in other words, G has a set V and G representing the graph)
Alternatively, $\deg(v) = |N_G(v)|$